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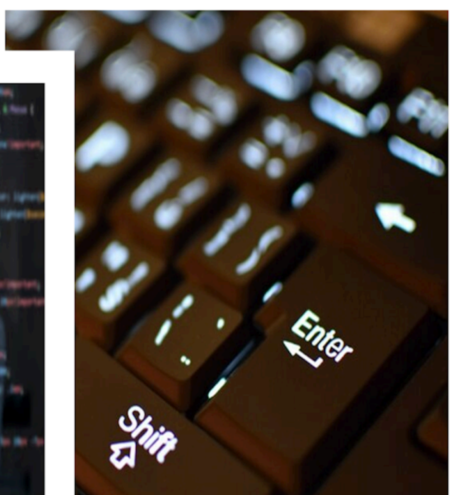
Computer Awareness

STUDY NOTES

Introduction to Computer



Introduction to Computers

Types of Computers

A computer is an electronic device that manipulates information, or data. It has the ability to store, retrieve, and process data. You may already know that you can use a computer to type documents, send email, play games, and browse the Web.

COMPUTER stands for Common Operating Machine Purposely Used for Technological and Educational Research.

Type	Specifications
PC (Personal Computer)	It can hold a single user at a time and this computer system has a moderately powerful microprocessor.
Workstation	It can also hold a single user at a time, similar to a personal computer, but it has a more powerful microprocessor.
Mini Computer	This is a multi-user system, i.e., capable of supporting even hundreds of users simultaneously.
Main Frame	This one is also a multi-user system. The software technology is different from the minicomputer.
Supercomputer	This is an extremely fast computer used to solve problems beyond human capabilities. This can execute hundreds of millions of instructions per second.

PC (Personal Computer)

- This can be defined as a small, relatively less expensive computer designed for an individual user. PCs are driven by the microprocessor chip that enables manufacturers to fabricate an entire CPU on one chip. Personals in businesses keep personal computers for desktop publishing, word processing, accounting , and for running spreadsheet and database management applications. Individuals owning it home, use it for playing games, surfing the Internet, etc.
- Even though personal computers are meant to be used as single-user

systems, these machines are normally linked together to create a network. If we consider operating power, presently premium models of the Mac and PC facilitate the equivalent computing power and graphics capability as compared to less expansive workstations from Sun Microsystems, Hewlett-Packard, and Dell.

Workstation

- It is a machine used for desktop publishing, software development, engineering applications (CAD/CAM), and other similar types of applications which needs a moderate amount of execution power and relatively high quality graphics capabilities.

- These machines generally come with a large, high-resolution graphics display, huge amounts of RAM, inbuilt network support, and a graphical user interface. Majority of workstations.
- It consists of a mass storage device like a disk drive, but a special type of workstation, known as diskless workstation, comes without a disk drive.
- Generally, UNIX and Windows NT are common operating systems for workstations. Similar to PC, workstations can also handle single-user but are typically connected together to form a local-area network, although they may operate as stand-alone systems.

Minicomputer

- It is a **medium size, multi-processing** machine capable of serving up to **250 users** simultaneously.

Mainframe

This is very large sized and expensive machine capable enough to support hundreds or even thousands of users at a time. Mainframe performs large quantity of programs concurrently and supports many executions of programs simultaneously.

Supercomputer

- Supercomputers are logically one of the fastest machines currently available. These are very expensive and are deployed for specialized operations that require an immense amount of mathematical calculations (number crunching).
- Few examples are - fluid dynamic calculations, weather forecasting, nuclear energy research, scientific simulations, (animated) graphics, electronic design, and analysis of geological data (e.g. in petrochemical prospecting).

Components of Computers

All types of computers are based on the same basic **logical structure** and execute the following **basic operations** for converting raw input data into useful information for their users.

1. **Take Input:** The process of feeding data and instructions to the computer.
2. **Store Data:** Saving unprocessed/processed data and instructions so that they are ready for processing as and when required.
3. **Processing Data:** Executing arithmetic and logical operations on the da- ta provided in order to transform it into useful information.
4. **Output Information:** Producing useful information/results for the user, like printed reports or visual information.

Input Unit

This part of the computer encapsulates devices with the help of which the user feeds data to the computer. It creates an interface between the user and the computer. The input devices transform the information into a form acceptable by the computer.

Output Unit

This part of the computer encapsulates devices with the help of which users receive the information from the computer. Output devices transform the output from the computer into a form understandable by the users.

Central Processing Unit (CPU)

- The brain of the computer is the Central Processing Unit. CPU executes all types of data processing functions. It saves data/intermediate results/instructions (program), and controls the operation of all parts of the computer.
- Following are the points to remember for Central Processing Unit (CPU):
    1. The CPU is taken as the brain of the computer.
    2. CPU facilitates all types of data processing operations.
    3. It saves data, intermediate results, and instructions (program).
    4. It handles the operating of all parts of the computer.
- The CPU itself has the following three components.
    1. Memory or Storage Unit
    2. Control Unit
    3. ALU (Arithmetic Logic Unit)

Memory or Storage Unit

- This part of the computer system works to store instructions, data, and intermediate results. This unit passes data to other parts of the computer when required. It is also referred to as an internal storage unit or most commonly, the main memory or the primary storage or Random Access Memory (RAM).
- It comes in various speeds, powers, and capability. Primary memory and secondary memory are two important types of memories used in the computer system. Responsibilities of the memory unit are –
    1. Works to store all the data and the instructions required for processing.
    2. Works to store intermediate results of processing.

3. Works to store the final results of processing before these results are forwarded to an output device.
4. All inputs and outputs are supplied through the main memory.

Control Unit

- This unit manages the operations of all parts of the computer but does not carry out any calculations or comparisons or actual data processing operations.
- Responsibilities of this unit are –
    1. For facilitating the transfer of data and instructions among other units of a system.
    2. It manages and coordinates all the units of the system.
    3. It receives the instructions from the memory, interprets them, and directs the operation of the system.
    4. It interacts with Input/output units to transfer data/results from storage.
    5. It does not perform processes or store data.

Arithmetic Logic Unit (ALU)

This unit consists of two subsections namely,

1. **Arithmetic Section**: Responsibility of the arithmetic unit is to execute arithmetic operations like addition, subtraction, multiplication, and division. A complete set of complex operations are executed by making iterative use of the above operations.
2. **Logic Section**: Responsibility of logic unit is to execute logic operations like NOR, AND, NOT, NAND, XOR, OR, etc.

Input Devices

Some of the Commonly used input units used in a computer system are follows:

1. Keyboard
2. Mouse
3. Joy Stick
4. Light pen
5. Track Ball
6. Scanner
7. Graphic Tablet
8. Microphone
9. Magnetic Ink Card Reader (MICR)
10. Optical Character Reader (OCR)
11. Bar Code Reader

12. Optical Mark Reader (OMR)

Keyboard

- A keyboard is the most basic, and very commonly used input device which helps to input data to the computer. The layout of the buttons in a normally used keyboard is similar to the traditional typewriter, but there are a few additional keys provided by different manufacturers for performing additional functions.
- Normally available keyboards in the market were of two sizes 84 keys and 101/102 keys, but now keyboards with 104 keys or 108 keys are also available for Windows and Internet.
- Following is the description of the buttons on the keyboard:

1. **Typing Keys**
    - ◊ These buttons include the letter keys (A-Z) and digit keys (09) which normally give the same layout like typewriters.
2. **Numeric Keypad**
    - ◊ These buttons are used to input the numeric data or cursor movement. Normally, it consists of a set of 17 keys that are laid out in the similar configuration used by most additional machines and calculators.
3. **Function Keys**
    - ◊ The 12 function keys are provided on the keyboard which are arranged in a row at the top of the keyboard. Each of these keys has a unique function and is used for some specific task.
4. **Control keys**
    - ◊ These buttons are used to cursor and screen control. It also adds four directional arrow keys. Following are also included in control keys: Home, End, Insert, Delete, Page Up, Page Down, Control (Ctrl), Alternate (Alt), Escape (Esc).
5. **Special Purpose Keys**
    - ◊ A regular keyboard also contains some special purpose buttons like Enter, Shift, Caps Lock, Num Lock, Space bar, Tab, and Print Screen.

Mouse

- It is the most commonly used pointing device. It is a very famous cursor-control device and the earlier versions of it were built over a small palm size box with a round ball at its base, which tracks the movement of the mouse and feeds digital signals to the CPU when the mouse buttons are pressed. Currently, mouse have a sensor at the bottom to detect cursor position.

- Basic versions if it has two buttons called the left and the right click button and a wheel is present between the buttons to provide scroll function. The movement of the mouse on a flat surface is used to control the position of the cursor on the display, but it cannot be used to feed text into the system directly.

Advantages

- ➔ Easy to use
- ➔ Not very expensive
- ➔ The cursor movement is faster than the arrow keys of the keyboard.

Joystick

- Just like a mouse, Joystick is also a pointing device, which is used to travel the cursor position on a display. It is a perpendicular stick having a spherical ball at its both lower and upper ends. The spherical ball connected at the circuit having sensors moved in a socket. The joystick controller can be traversed in all directions.

- The functionality of the joystick is just like a mouse. Its applications are generally in Computer Aided Designing (CAD) and playing computer games.

Light Pen

- Just like a pen, a light pen is a digital pointing device. This device is used to select an item displayed in the menu or to draw anything on the display. It contains a photocell and an optical sensor placed in a small tube at the tip.
- When the light pen tip is travelled on the display, its photocell sensing element calculates the screen location and sends the digital signal to the CPU.

Track Ball

- A track ball is another type of input device similar to the mouse. It is mostly helpful with a notebook or laptop computer, instead of a mouse. It consists of a ball on the top which is half inserted and connected to the sensors. By moving fingers on the ball, the cursor can be moved.

- Because the whole device is not moved to move the cursor, a track ball requires less space as compared to the mouse. This device comes in different shapes like a ball, a button, or a square.

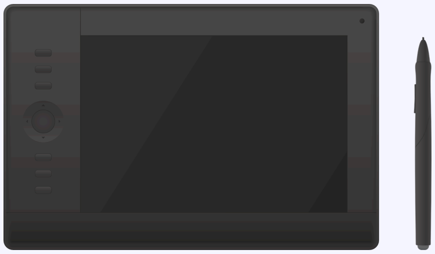
Scanner

- A Scanner is another input device, which works very much like a photocopy machine. It can be used when some information on paper is to be transferred to the hard disk of the computer in digital format for further manipulation.

- Scanner captures high resolution images from the source which are then converted into a digital format that can be saved on the disk and shared digitally. These digital images can be edited before they are printed.

Digitizer

- Digitizer is also an input device which is used to convert analog information into digital form. Using a digitizer, signals from the television or camera can be converted into a series of numbers that could be saved in a computer hard drive. They can be helpful with the computer to create a picture of whatever the camera has been pointed at.
- Digitizer is also called Tablet or Graphics Tablet because it transforms graphics and pictorial data into binary inputs to the system. A special kind of graphic tablet as a digitizer is used for fine works of drawing and image related applications.



Microphone

- Microphone is a very common input device used to input sound that is then stored in a digital form.
- Nowadays microphones are used in almost every industry and devices related to sound recording or

transmission.
- The microphone is used for different applications like adding sound to a multimedia presentation or for mixing music.

Magnetic Ink Card Reader (MICR)

- MICR input device is commonly seen in banks as there are huge amounts of cheques to be processed every day. The important details like bank's code number and cheque number are printed on the cheques with a specially designed ink based on the particles of magnetic material that are readable by the sensors of the machine.

- This process of reading is known as Magnetic Ink Character Recognition (MICR). The main benefits of using MICR is that it is comparatively fast and less prone to errors.

Optical Character Reader (OCR)

- It is an input device which is used to read a printed text similar to the scanner, but the method of reading and the type of output generated is different.
- The format to be fed to the OCR is predefined and can't be used like an ordinary scanner.
- OCR, unlike scanner, scans the text optically, character by character, converts them into a machine readable digital code instead of creating high resolution images, and stores the text on the system memory.

Bar Code Readers

- Bar Code Reader is an input device used to read special format bar coded data (data in the form of light and dark lines). Bar coded data is commonly used to create price tags, in labelling goods, numbering the books, etc. It can be a handheld scanner or can be embedded in a stationary scanner on the table top.

- A flash of light from the device strikes to the surface of the object and reflects back to the sensors present behind the source of light to collect the input.
- Bar Code Reader can only scan a bar code image, converts it into an alphanumeric value, which is then transferred to the computer that the bar code reader is connected to.

Optical Mark Readers (OMR)

- OMR is a special type of input machine. This kind of optical scanner is used to recognize the type of mark made by pen or pencil. It is used where one out of a predefined alternative is to be selected and marked.
- The format to be fed to the OMR is predefined and can't be used like an ordinary scanner.
- It is specially used for automation of the answer sheet analysis of examinations having multiple choice questions.

Output Devices

Following are some of the commonly used output devices used in a computer system:

1. Monitors
2. Printer

Monitors

- Monitors or Visual Display Unit (VDU), are the primary output device of a computer system. It creates images from tiny dots, so called pixels that are arranged in a well- defined rectangular form. The depth and sharpness of the image depends upon the number and size of the pixels.
- There are two types of digital screen used for monitors:
  1. Cathode-Ray Tube (CRT)
  2. Flat-Panel Display

Cathode-Ray Tube (CRT)

- The CRT display consists of small picture elements known as pixels. The tinier the pixels, the greater the image clarity or resolution. It requires a very large number of pixels emitting light to create a character, just like the letter ‘e’ in the word help.
- A limited number of characters can be arranged on a screen at a time. The screen can be categorized into a series of character boxes - fixed places on

the screen where a standard character can be shown.
- The capability of earlier screens was to display 80 characters of data horizontally and 25 lines vertically.
- There are some disadvantages of CRT –
    1. Large in Size
    2. High power consumption.

Flat-Panel Display Monitor

- The flat-panel display is a class of video output devices that have decreased volume, weight and power requirement as compared to the CRT. It is possible to hang them on walls or wear them on wrists. In everyday evolving technology, uses of flat- panel displays include calculators, video games, monitors, laptop computers, and graphics displays.
- The flat-panel display is categorized into following two categories –
    1. **Emissive Displays** – Emissive displays

       panels are output devices that transform electrical energy into light. For instance, plasma panel and LED (Light-Emitting Diodes).
    2. **Non-Emissive Displays** – Non-emissive displays are based on optical

       effects to transform sunlight or light from some other source into graphics patterns. For example, LCD (Liquid-Crystal Device).

Printers

- A printer is a very commonly used output device, which is used to print information on paper.
- There are two types of printers –
    1. Impact Printers
    2. Non-Impact Printers

Impact Printers

- Impact printers print by hitting the characters on the ribbon containing ink, which is then pressed on the paper.
- Following are the characteristics of Impact Printers –
    1. Very low consumable costs
    2. Very noisy

3. Useful for bulk printing due to low cost
4. There is physical contact of the embossed characters with the paper to produce an image

- These printers are of two types –
    1. Character printers
    2. Line printers

Character Printers

- Character printers are the type of printers that are capable of printing only one character at a time.
- These can be further categorized into two types:
    1. Dot Matrix Printer (DMP)
    2. Daisy Wheel

Dot Matrix Printers

- In the earlier days, one of the most commonly used printers was Dot Matrix Printer. These printers were used most commonly because of their ease of usage, printing and affordable printing price. Every character to be printed on paper is created by rearranging matrix of metallic pins of size (5x7, 7x9, 9x7 or 9x9) at very fast speed which come out to create a character which is why it is known as Dot Matrix Printer.
- Advantages:
    ◊ Inexpensive
    ◊ Widely Used
    ◊ Other language characters can be printed
- Disadvantages:
    ◊ Slow Speed
    ◊ Too much heat generation
    ◊ Poor Quality

Daisy Wheel

- The head is fixed on a wheel and pins creating characters are like petals of Daisy (flower) that is why it is known as Daisy Wheel Printer.
- These printers are commonly used for word- processing tasks in offices that need a very less number of letters to be sent here and there with very nice quality.

- Advantages:
    - ◊ More reliable than DMP
    - ◊ Better quality
    - ◊ Fonts of character can be easily changed
- Disadvantages:
    - ◊ Slower than DMP
    - ◊ Noisy
    - ◊ More expensive than DMP

Line Printers

- These printers are capable of printing one line at a time.
- These are of two types –
    1. Drum Printer
    2. Chain Printer

Drum Printers

- The head of this printer is in the shape of a drum and that is why it is known as a drum printer. The printing drum surface is divided into a number of tracks which are equivalent to the size of the paper.
- For instance, if a paper is of width of 200 characters, then the drum will have 200 tracks. A fixed character set is embossed on the track which cannot be altered by any means.
- To use a different character set, a different drum needs to be purchased like the one with 48 character set, or 64 and 96 characters set.
- A single rotation of the drum containing permanent characters prints a single line. These printers offer fast printing speed and can print 300 to 2000 lines per minute.
- Advantages:
    - ◊ Very high speed
- Disadvantages:
    - ◊ Very expensive
    - ◊ Characters fonts cannot be changed
    - ◊ obsolete

Chain Printers

- In this type printer, a chain consisting of character set is used, that is why it is known as Chain Printer. The predefined-standard character set can have 48, 64, or 96 characters.

- Advantages:
    - ◊ It is easier to change character fonts.
    - ◊ Different languages can be used by replacing chains with the same printer.
- Disadvantages:
    - ◊ Noisy

Non-Impact Printers

- Non-impact printers do not hit the paper with any embossed character-containing instrument. These printers are able to print a complete page all at once, this is the reason why they are also known as Page Printers.
- These printers are of two types –
    - ◊ Laser Printers
    - ◊ Inkjet Printers
- Characteristics of Non-impact Printers
    - ◊ Faster than impact printers
    - ◊ They are not noisy
    - ◊ High quality
    - ◊ Supports many fonts and different character size.

Chain Printers

- These are very commonly used these days and are a type of non-impact page printers. They use laser lights to charge the metallic drum in the shape of characters or objects to be printed which attracts powdered ink. Then this drum is pressed over the page to create printed output.

- Advantages
    - ◊ Very high speed
    - ◊ Very high quality output
    - ◊ Good graphics quality
    - ◊ Supports many fonts and different character size.
- Disadvantages
    - ◊ Expensive
    - ◊ Produces heat
    - ◊ Difficult to maintain

Inkjet Printers

- Inkjet printers are also very commonly used printers these days. They use ink cartridges to print characters by spraying small drops of ink in a very precise manner onto the paper. Inkjet printers are very versatile and can produce relatively high quality output with presentable features.
- They make very less noise as compared to other printers because no hammering is done and these have many different modes of printing. Using these printers, color printing is also possible. Some advanced and newer models of Inkjet printers are capable enough to produce multiple number copies of printing also.
- Advantages
    - ◊ High quality printing
    - ◊ More reliable
- Disadvantages
    - ◊ Expensive as the cost per page is high
    - ◊ Slow as compared to laser printer

Booting

- Starting a computer or a computer-embedded device is called **booting**. Booting takes place in two steps –
    - Switching on power supply
    - Loading operating system into computer's main memory
    - Keeping all applications in a state of readiness in case needed by the user
- The first program or set of instructions that run when the computer is switched on is called **BIOS** or **Basic Input Output System**. BIOS is a **firmware**, i.e. a piece of software permanently programmed into the hardware.

- If a system is already running but needs to be restarted, it is called **rebooting**. Rebooting may be required if a software or hardware has been installed or the system is unusually slow.
- There are two types of booting –
    - **Cold Booting** – When the system is started by switching on the power supply it is called cold booting. The next step in cold booting is loading of BIOS.
    - **Warm Booting** – When the system is already running and needs to be restarted or rebooted, it is called warm booting. Warm booting is faster than cold booting because BIOS is not reloaded.